MTH101 FINAL TERM PREPARATION BY BRAINY SQUAD 80+ MCQ'S

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Question No: 1

Let $f(x)$ is a function such that as x approaches a real number a, either from left or right-hand-side, the function values increases or decreases unboundedly then

$\lim_{x \to a} f(x)$

- ► Exist
- ► Does not exist

Question No: 2

$\frac{d(\sec x)}{dx}$ =

- $(\sec x)(\tan x)$
- $(\sec x)(\tan x)$
- $(co\sec x)(co\ t\ x)$
- $(co\sec x)(\tan x)$

Question No: 3

Consider a function $h(x)$ and a constant c then $\frac{d}{dx}((c)\{h(x)\})$ = _____

- 0
- $\frac{d}{dx}(h(x))$
- $\frac{d}{dx}(h(cx))$
- $c\frac{d}{dx}(h(x))$

Queion No: 4 _____

$-\infty, +\infty)$ $\lim_{x\to-\infty}$ then has _____ on

- maximum value but no minimum
- minimum value but no maximum

If f is continuous function such that

f

(

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$f(x) = +\infty$ *and* $\lim_{x \to +\infty} f(x) = +\infty$

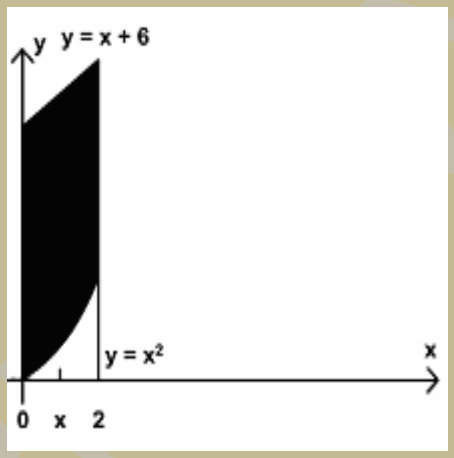
- both maximum and minimum value

Question No: 5

Sigma notation is represented by which of the following Greek letter?

- χ
- η
- Σ
- ψ

Question No: 6 (Marks: 1) - Please choose one



In the following figure, the area enclosed is bounded below by :

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- $y = x + 6$
- $y = x^2$
- $x = 2$
- $x = 0$

Question No: 7 (Marks: 1)

At what points the two curves: $y = x^2 \text{ and } y = x + 6$ intersect ?

- $x = 0 \text{ and } x = 2$
- $x = 0 \text{ and } x = 3$
- $x = 2 \text{ and } x = 3$
- $x = -2 \text{ and } x = 3$

Question No: 8 (Marks: 1)

Let the solid generated by the region enclosed between $y = \sqrt{x}$; $x = 1, x = 4$ and the x-axis is revolved about the y-axis. Which of

- $V = \int_1^4 2\pi x \sqrt{x} dx$

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Question No: 9

Let f is a smooth curve on the interval [a, b]. What is the arc length L of the curve f(x) defined over the interval [a, b]?

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))}$$

$$L = \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))\Delta x_k}$$

$$L = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1 + (f'(x^*_k))^2 \Delta x_k}$$

$$L = \sum_{k=1}^{n} \sqrt{1 + (f(x^*_k))\Delta x}$$

Question No: 10

For a graph to be symmetric about y-axis means, for each point (x,y) on the graph,the point---------- is also on the graph

- (x , -y)
- (-x , y)

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- (-x , -y)

Question No: 11

The graph $x = y^2$ is symmetric about --------- axis

- X-axis
- Y-axis
- Origin

Question No: 12

If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y , we say that y is .....of x

- relation
- function
- not a function
- not a relation

Question No: 13

Domain of the function $y = \frac{(x^2 - 4)}{(x - 2)}$ is

- $(-\infty, 2) U (2, +\infty)$
- $(-\infty, 2)$
- $(-\infty, 0)$

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Question No: 14 (Marks: 1) - Please choose one

Tan(x) is continuous every where except at points

- $\pm \frac{k\pi}{2} (k = 1, 3, 5, ...)$
- $\pm \frac{k\pi}{2} (k = 2, 4, 6, ...)$

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Question No: 15

$\lim_{x \to 0} \frac{\sin x}{x}$ = ----------

- -1
- 2
- 0
- 1

Question No: 16

How the series $1-3+5-7+9-11$ can be expressed in sigma notation?

- $\sum_{k=0}^{k=5}(-1)^k(2k+1)$
- $\sum_{k=1}^{k=5}(-1)^k(2k+1)$
- $\sum_{k=1}^{k=5}(2k+1)$
- $\sum_{k=1}^{k=5}(2k+1)$

Question No: 17 (Marks: 1) - Please choose one

Let the region bounded by the curve $y=\sqrt[3]{x}$, the x-axis, and the line $x=3$ is revolved about the y-axis to generate a solid. Which of the following equation gives the volume of a solid by cylindrical shells?

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3

$$V = \int_0^3 2\pi x\sqrt{x}\,dx$$

▶

$$V = \int_0^3 x\sqrt{x}\,dx$$

▶

Question No: 18 (Marks: 1) - Please choose one

Let $y = \frac{2\sqrt{2}}{3}x^{\frac{3}{2}};\ 0 \le x \le 2$ then which of the following is the length of the curve?

$$L = \int_0^2 \sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L = \int \sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L = \int_0^2 \sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]^2}\,dx$$

▶

$$L = \int_0^2 \sqrt{1+\left[\frac{d}{dx}\left(\frac{2\sqrt{2}}{3}x^{\frac{3}{2}}\right)\right]}$$

▶

Question No: 19 (Marks: 1) - Please choose one

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$\frac{2}{3}$ is known as

- An even number
- Irrational Number
- A natural Number
- Rational Number

Question No: 20 (Marks: 1) - Please choose one

For a function *f*, let $f'(x_n) = 0 \quad for\ some\ n$.

Does the Newton's Method works for approximating the solution of $f(x) = 0$?

- Yes
- No

Question No: 21 (Marks: 1) - Please choose one

The Mean Value Theorem states that "Let function *f* be differentiable on (a,b) and continuous on [a, b], then there exist at least one point c in (a,b) where ..... "

- $f'(c) = \frac{f(b) - f(a)}{b - a}$
- $f(c) = \frac{f(b) - f(a)}{b - a}$
- $f(c) = \frac{f(a) - f(b)}{b - a}$
- $f'(c) = \frac{f(a) - f(b)}{b - a}$
-

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Question No: 22 (Marks: 1) - Please choose one

If there is some function F such that $\frac{d}{dx}[F(x)] = f(x)$ then any function of the form $F(x) + C$ is----------of $f(x)$

- Derivative
- Antiderivative
- Slope
- Maximum value

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Question No: 23 (Marks: 1) - Please choose one

If *f* and *g* are continues function on an interval [a, b]
and $f(x) \geq g(x) \quad for \quad a \leq x \leq b$, then area is bounded by the lines parallel to:

- ► X -axis
- ► Y-axis
- ► Both X -axis and Y-axis

Question No: 24 (Marks: 1) - Please choose one

What is the sum of following series?

$1^3 + 2^3 + 3^3 + 4^3 + _____ + n^3$

- ► $\frac{n(2n)(2n+1)}{6}$
- ► $\frac{(n+1)(n+2)}{2}$
- ► $\left[\frac{n(n+2)}{2}\right]^2$
- ► $\left[\frac{n(n+1)}{2}\right]^2$

Question No: 25 (Marks: 1) - Please choose one

$\frac{5}{7}\times 1^2 + \frac{5}{7}\times 2^2 + \frac{5}{7}\times 3^2 + \frac{5}{7}\times 4^2 .... + \frac{5}{7}\times n^2 = _____$

- ► $\frac{5n(n+1)(2n+1)}{42}$

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$\frac{5n(n+1)}{14}$

► $\frac{5n^2(n+1)^2}{14}$

► $\frac{5(n+1)(2n+1)}{42}$

►

Question No: 26 (Marks: 1) - Please choose one

$\int_a^a f(x)dx =$ _____

If point a is in the domain of function f, then

► $f'(x)$

► $f(x)$

► 0

► 1

Question No: 27 (Marks: 1) - Please choose one

If $a_1 > a_2 > a_3 > ..... > a_n > ....$, then a sequence $\{a_n\}$ is .....

► Increasing

► Nondecreasing

► Decreasing

► Nonincreasing

Question No: 28 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the difference between successive terms $a_{n+1} - a_n \leq 0$ then the sequence is known as:

► Increasing

► Decreasing

► Nondecreasing

► Nonincreasing

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Question No: 29 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} < 1$ then the sequence is known as:

- Increasing
- Decreasing
- Nondecreasing
- Nonincreasing

Question No: 30 (Marks: 1) - Please choose one

For a sequence $\{a_n\}$ if the ratio of successive terms $\frac{a_{n+1}}{a_n} \geq 1$ then the sequence is known as :

- Increasing
- Decreasing
- Nondecreasing
- Nonincreasing

Question No: 31 (Marks: 1) - Please choose one

Which of the following option is true for the sequence $a_n = \left\{\frac{1}{n}\right\}^{\infty}$?

- Increasing
- Decreasing
- Nonincreasing
- Nondecreasing

Question No: 32 (Marks: 1) - Please choose one

If the partial sum of a series is finite then the series will/will be:

- Divergent
- Convergent

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► Give no information

Question No: 33 (Marks: 1) - Please choose one

If the geometric series $a + ar + ar^2 + ar^3 + \ldots + ar^{k-1} + \ldots \quad where\ (a \neq 0)$,

$|r| < 1$

then which of the following is true for the given series?

- Converges
- Diverges
- Gives no information
-

Question No: 34 (Marks: 1) - Please choose one

$\rho = \lim_{k \to +\infty} \frac{u_{k+1}}{u_k}$

If where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be ..... ?

- Convergent
- Divergent
- Give no information

Question No: 35 (Marks: 1) - Please choose one

$\rho = \lim_{k \to +\infty} \sqrt[k]{u_k}$

If where $\rho > 1$ then the series $\sum u_k$ with positive terms will /will be ..... ?

- Convergent
- Divergent
- Give no information

Question No: 36 (Marks: 1) - Please choose one

In alternating series test, which one of the following condition must be satisfied?

- $\lim_{k \to \infty} a_k = 1$
- $a_1 > a_2 > a_3 \ldots\ldots > a_k > \ldots.$
- $a_1 \leq a_2 \leq a_3 \ldots\ldots \leq a_k \leq \ldots.$
-

Question No: 37 (Marks: 1) - Please choose one

A series of the form

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$\sum_{k=1}^{\infty} (-1)^n a_k$ is called _____.

- Alternating series
- Geometric series
- Arithmetic series
- Harmonic series

Question No: 38 (Marks: 1) - Please choose one

Which of the following is the Maclaurin series for e^x ?

- $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$
- $x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$
- $1+x+\frac{x^3}{3!}+\ldots+\frac{x^k}{k!}+\ldots$
- $1-x+\frac{x^3}{3!}-\ldots-\frac{x^k}{k!}-\ldots$
-

Question No: 39 (Marks: 1) - Please choose one

Which of the following is the work done *W* if an object moves in the positive direction along a coordinate line while subject to a force *F(x)* in the direction of motion over an interval [0,3]?

- $W=\int_2^3 3x\,dx$
- $W=\int_0^3 3x\,dx$
- $W=\int_0^3 F(x)\,dx$

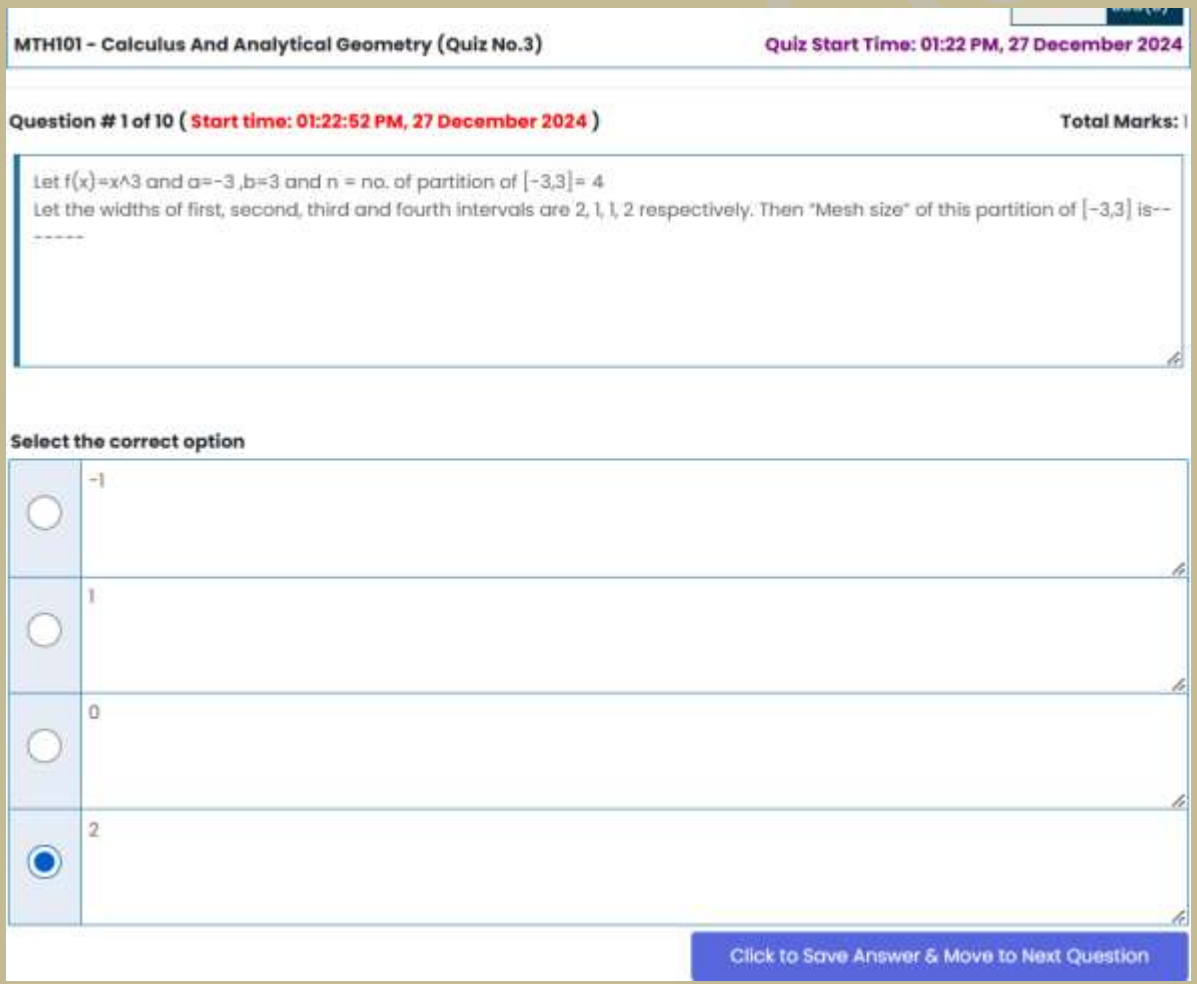
$$W = \int_{3}^{0} F(x)dx$$

▶

Question No: 40 (Marks: 1) - Please choose one

Which of the following is the spring constant *k* if a spring whose natural length is 2*m* exerts a force of 3*N* when stretched 1*m* beyond its natural length?

- ▶ 3 *x*
- ▶ 3 *N/m*
- ▶ 2 *m*
- ▶ 3 *m/N*



MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 2 of 10 (Start time: 01:23:46 PM, 27 December 2024) Total Marks:

Summation of 2 where sum ranges from 0 to 10 equals 20.

Select the correct option

- True
- False

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 3 of 10 (Start time: 01:24:27 PM, 27 December 2024) Total Marks: 1

1+2+3+...+339 equals _____

Select the correct option

Reload Math Equations

- 338(340)/6
- 338(339)/6
- 338(340)/2
- 339(340)/2

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 4 of 10 (Start time: 01:25:15 PM, 27 December 2024) Total Marks: 1

If f(x) = x^5 + 6, then which of the following is Not true about it.

Select the correct option

- (●) Its anti – derivative is x^6/6 + 6x + 10.
- ( ) Its anti – derivative is x^6/6 + 12x
- ( ) Its anti – derivative is x^6/6 + 6x + 6.
- ( ) Its anti – derivative is x^6/6 + 6x.

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 5 of 10 (Start time: 01:26:38 PM, 27 December 2024) Total Marks: 1

If the closed interval [-10,x] is divided into '20' equally spaced subintervals each of which having the width equals to '1' unit then the value of 'x' is --------

Select the correct option

- 30
- 0
- 20
- 10

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 6 of 10 (Start time: 01:27:22 PM, 27 December 2024) Total Marks:

The integral $\int \frac{10x}{(5x^2+1)^2}\,dx$ will be equal to ?

Select the correct option

Reload Math Equations

- $\frac{1}{5x^2+1}+c$
- $-\frac{1}{(5x^2+1)^3}+c$
- $-\frac{1}{5x^2+1}+c$
- $-\frac{1}{5x+1}+c$

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:22 PM, 27 December 2024

Question # 7 of 10 (Start time: 01:28:10 PM, 27 December 2024) Total Marks: 1

Sigma notation is used to write lengthy.....in compact form.

Select the correct option

- difference
- products
- quotient
- sums

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 01:22 PM, 27 December 2024**

Question # 8 of 10 (Start time: 01:28:42 PM, 27 December 2024) **Total Marks:**

What is the estimated area under $f(x) = 9 - x^2$ from $x = 0$ to $x = 4$ with mid points for $n = 1$?

Select the correct option

Reload Math Equations

- 21
- 24
- 20 (selected)
- 28

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 01:22 PM, 27 December 2024**

Question # 9 of 10 (Start time: 01:29:45 PM, 27 December 2024) **Total Marks:**

For the area under the curve $f(x) = 2x$ from $x = 0$ to $x = 12$ with left points approximations for $n = 3$, what will be the values of x_k^* ?

Select the correct option

Reload Math Equations

- 0, 3 and 6
- 0, 5 and 10
- 0, 4 and 8 (selected)
- 0, 6 and 12

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 01:22 PM, 27 December 2024**

Question # 10 of 10 (Start time: 01:30:50 PM, 27 December 2024) **Total Marks:** 1

Integral of (5/2)cos(x) is

Select the correct option

- (3/2)csc(x)+C
- (5/2)tan(x)+C
- (5/2)sin(x)+C
- None of these

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 1 of 10 (Start time: 01:34:30 PM, 27 December 2024) Total Marks:

To get better approximation to actual area under a continuous curve over a closed interval, we have to increase .....

Select the correct option

- Size of the interval
- Width of the subintervals
- Number of subintervals
- Total area

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 01:34 PM, 27 December 2024**

Question # 2 of 10 (Start time: 01:35:25 PM, 27 December 2024) **Total Marks: 1**

The derivative of the area under the continuous function f(x)= 2+3Sinx in the interval[-pi,pi] is---------

Select the correct option

- (●) 2+3Sinx
- ( ) 2+3Cosx
- ( ) 2-3Cosx
- ( ) 2-3Sinx

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 3 of 10 (Start time: 01:36:10 PM, 27 December 2024) Total Marks:

Summation of 2 where sum ranges from 0 to 10 equals 20.

Select the correct option

- (●) False
- ( ) True

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 4 of 10 (Start time: 01:36:43 PM, 27 December 2024) Total Marks:

The integral $\int \sec^2(2x^2)\,.4x\,dx$ will be equal to ?

Select the correct option

Reload Math Equations

- ( ) $\sec^2(2x) + c$
- ( ) $\tan(2x) + c$
- ( ) $\sec(2x^2).\tan(2x^2) + c$
- (●) $\tan(2x^2) + c$

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 5 of 10 (Start time: 01:37:17 PM, 27 December 2024) Total Marks: 1

The estimated area under $f(x) = 12 / x$ from $x = 1$ to $x = 3$ with right end points for $n = 2$ is _____.

Select the correct option

- 18.
- 10
- None of these.
- 20.

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 01:34 PM, 27 December 2024**

Question # 6 of 10 (Start time: 01:38:10 PM, 27 December 2024) **Total Marks:** 1

Let f(x)=x^3 and a=-3 ,b=3 and n = no. of partition of [-3,3]= 4
Let the widths of first, second, third and fourth intervals are 2, 1, 1, 2 respectively. Then "Mesh size" of this partition of [-3,3] is---

Select the correct option

- 0
- -1
- 1
- 2

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 7 of 10 (Start time: 01:38:47 PM, 27 December 2024) Total Marks: 1

$$\int \tan x dx = _____.$$

Select the correct option

Reload Math Equations

- $\sec^2 x + C$
- (•) $\ln|\sec x| + C$
- $\ln|\sin x| + C$
- $\ln|\cos x| + C$

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 8 of 10 (Start time: 01:39:24 PM, 27 December 2024) Total Marks: 1

The Area A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles:

Select the correct option

Reload Math Equations

- $A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \Delta x \left[f(x_1) + f(x_2) + \ldots + f(x_n)\right]$
- (•) $A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \left[f(x_1)\Delta x + f(x_2)\Delta x + \ldots + f(x_{n-1})\Delta x\right]$
- $A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \frac{1}{\Delta x} \left[f(x_1) + f(x_2) + \ldots + f(x_n)\right]$
- $A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \left[f(x_1) + f(x_2) + \ldots + f(x_n)\right]$

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 9 of 10 (Start time: 01:40:16 PM, 27 December 2024) Total Marks: 1

$3(1^2)+3(2^2)+3(3^2)+_+3(15^2)$ equals _____.

Select the correct option

Reload Math Equations

- ○ 15(16)(29)/2
- ◉ 15(16)(31)/2
- ○ 15(16)(17)/2
- ○ 15(16)(30)/2

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 01:34 PM, 27 December 2024

Question # 10 of 10 (Start time: 01:41:05 PM, 27 December 2024) Total Marks: 1

summation of (n+i);where (i varies from 1 to 1) =.....

Select the correct option

- ◉ n+1
- ○ n
- ○ 1
- ○ 0

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 1 of 10 (Start time: 07:28:43 AM, 28 December 2024) Total Marks: 1

Summation of 'kx' where k goes from 1 to 5 equals

Select the correct option

- None of these
- 15k
- 15x
- 55

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 07:28 AM, 28 December 2024**

Question # 2 of 10 (Start time: 07:29:32 AM, 28 December 2024) **Total Marks:**

If a function $f(x)$ is defined on an interval
$[a, b]$ satisfying the conditions of the Rolle's Theorem, then which of the following is true about it.

Select the correct option

- It must have both absolute maximum and minimum values
- None of these.
- It has only absolute maximum value
- It has only absolute minimum value.

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 3 of 10 (Start time: 07:30:49 AM, 28 December 2024) Total Marks: 1

Subdivide the interval [1, 5] into 'n' equally spaced subintervals then the width of each sub-interval is ------

Select the correct option

- 5
- 1/n
- 5/n
- 4/n

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 4 of 10 (Start time: 07:31:33 AM, 28 December 2024) Total Marks: 1

The integral $\int (x^3+1)^{10}.3x^2\,dx$ will be equal to ?

Select the correct option

Reload Math Equations

- $-\frac{(x^3+1)^{11}}{11}+c$
- $\frac{(x^3+1)^{11}}{11}+c$
- None of these
- $\frac{(x^3-1)^{11}}{11}+c$

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 5 of 10 (Start time: 07:32:14 AM, 28 December 2024) Total Marks:

For Rolle's Theorem, f is differentiable on the interval _____.

Select the correct option

Reload Math Equations

- $[a, b)$
- $(a, b]$
- $[a, b]$
- (a, b)

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 6 of 10 (Start time: 07:32:49 AM, 28 December 2024) Total Marks:

The indefinite integral of 'sec(x)tan(x)' is_____

Select the correct option

- Cotx +c
- Secx+c
- Tanx+c
- Sinx+c

Click to Save Answer & Move to Next Question

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sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 7 of 10 (Start time: 07:33:42 AM, 28 December 2024) Total Marks: 1

In the notation: $\int_a^b f(x)dx$, $f(x)$ is called_____.

Select the correct option

Reload Math Equations

- (●) Integrand
- ( ) Integration
- ( ) Differential
- ( ) None of these

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) Quiz Start Time: 07:28 AM, 28 December 2024

Question # 8 of 10 (Start time: 07:34:16 AM, 28 December 2024) Total Marks:

'x.ln(x/e)' is the anti-derivative of 'lnx' if and only if the derivative of 'x.ln(x/e)' is euqal to --------

Select the correct option

- (●) lnx
- ( ) x
- ( ) ln(x/e)
- ( ) xln(x/e)

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No.3) **Quiz Start Time: 07:28 AM, 28 December 2024**

Question # 9 of 10 (Start time: 07:35:29 AM, 28 December 2024) **Total Marks:** 1

Let A be the area of a rectangle under a continuous function f(x) over a closed interval [a, b]. If this area is divided in to 'n' sub-rectangles then width of each approximated sub-intervals is ---------

Select the correct option

- (a-b)/2
- (a-b)/n
- (b-a)/2n
- (b-a)/n

Click to Save Answer & Move to Next Question

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